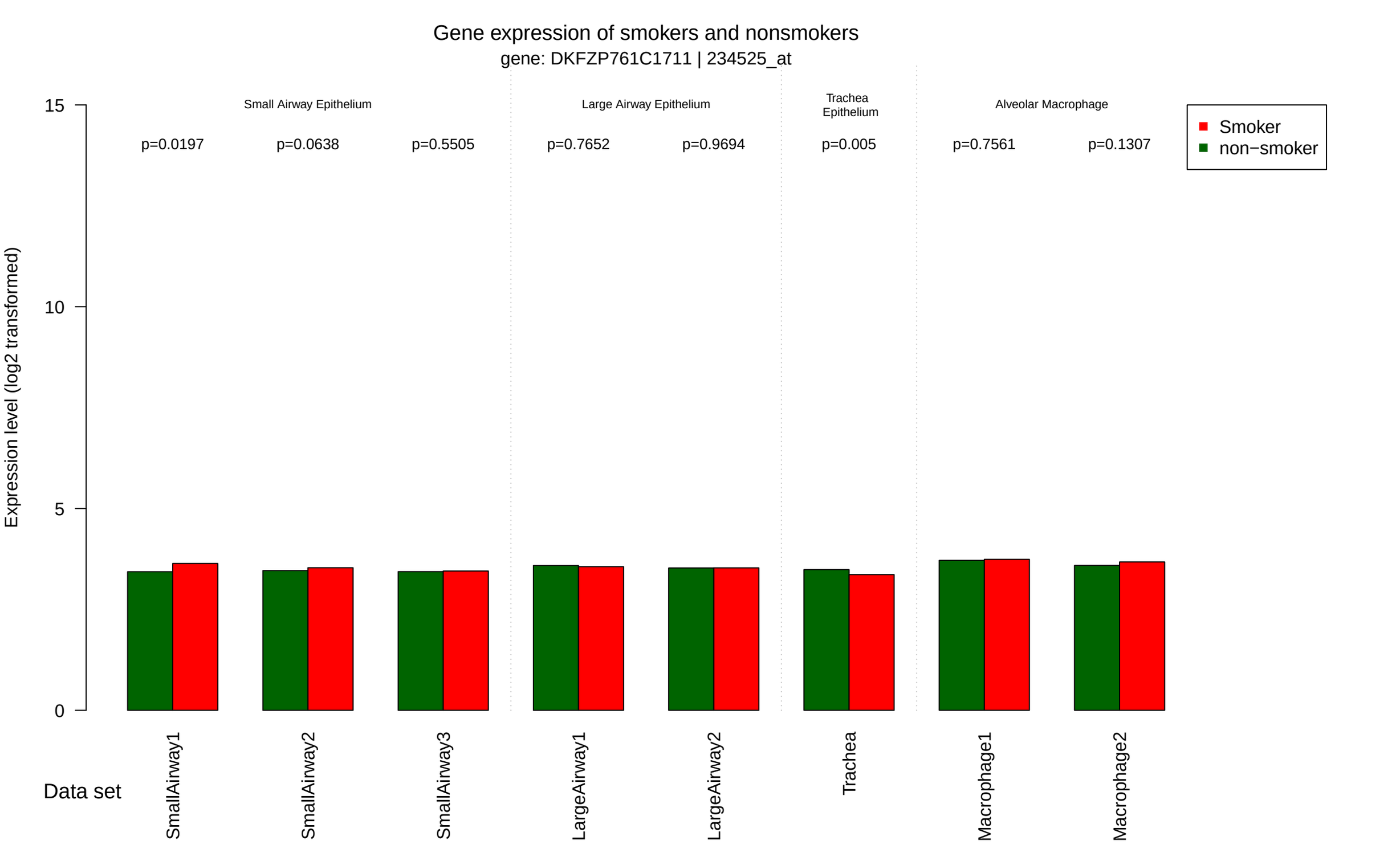
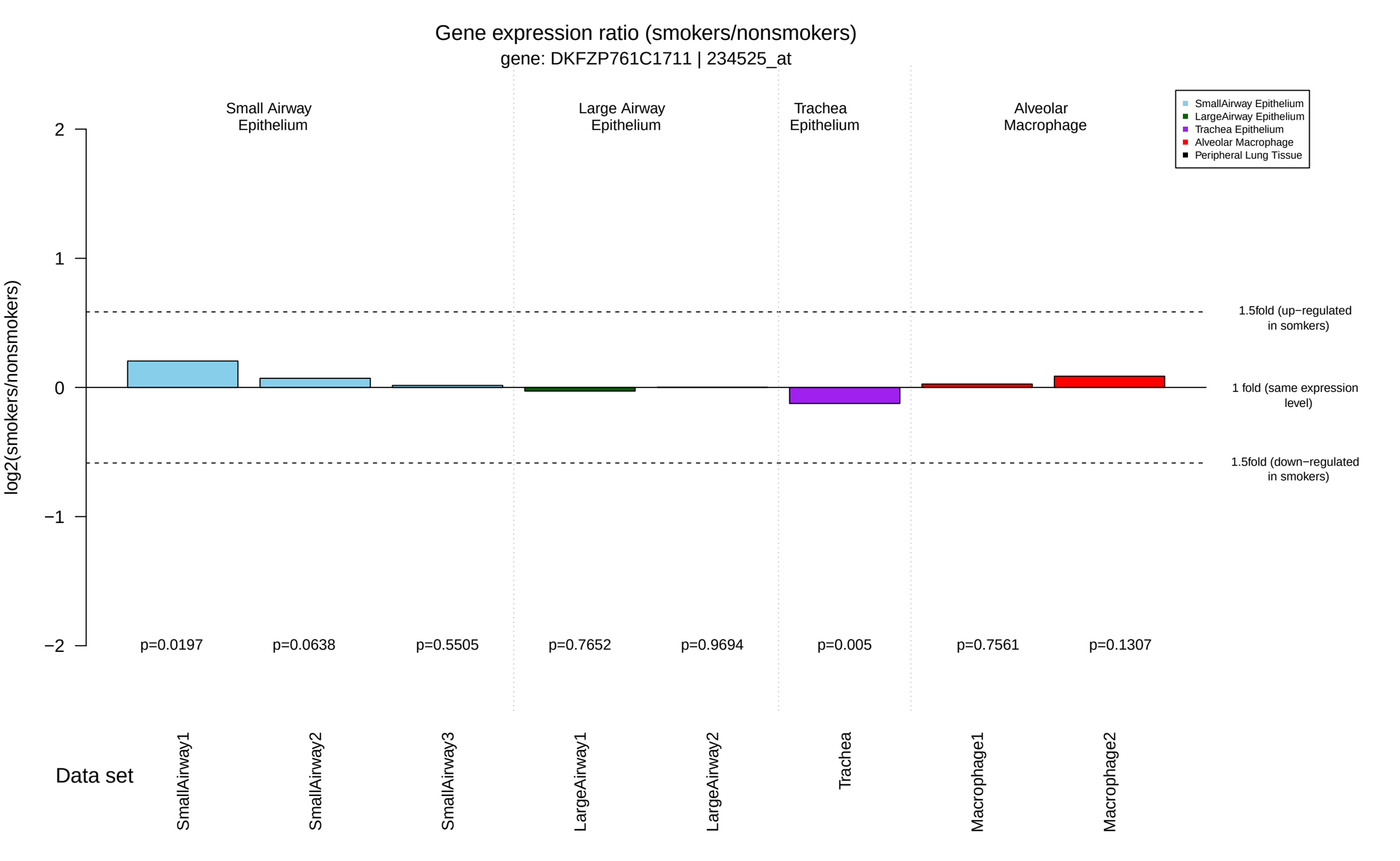
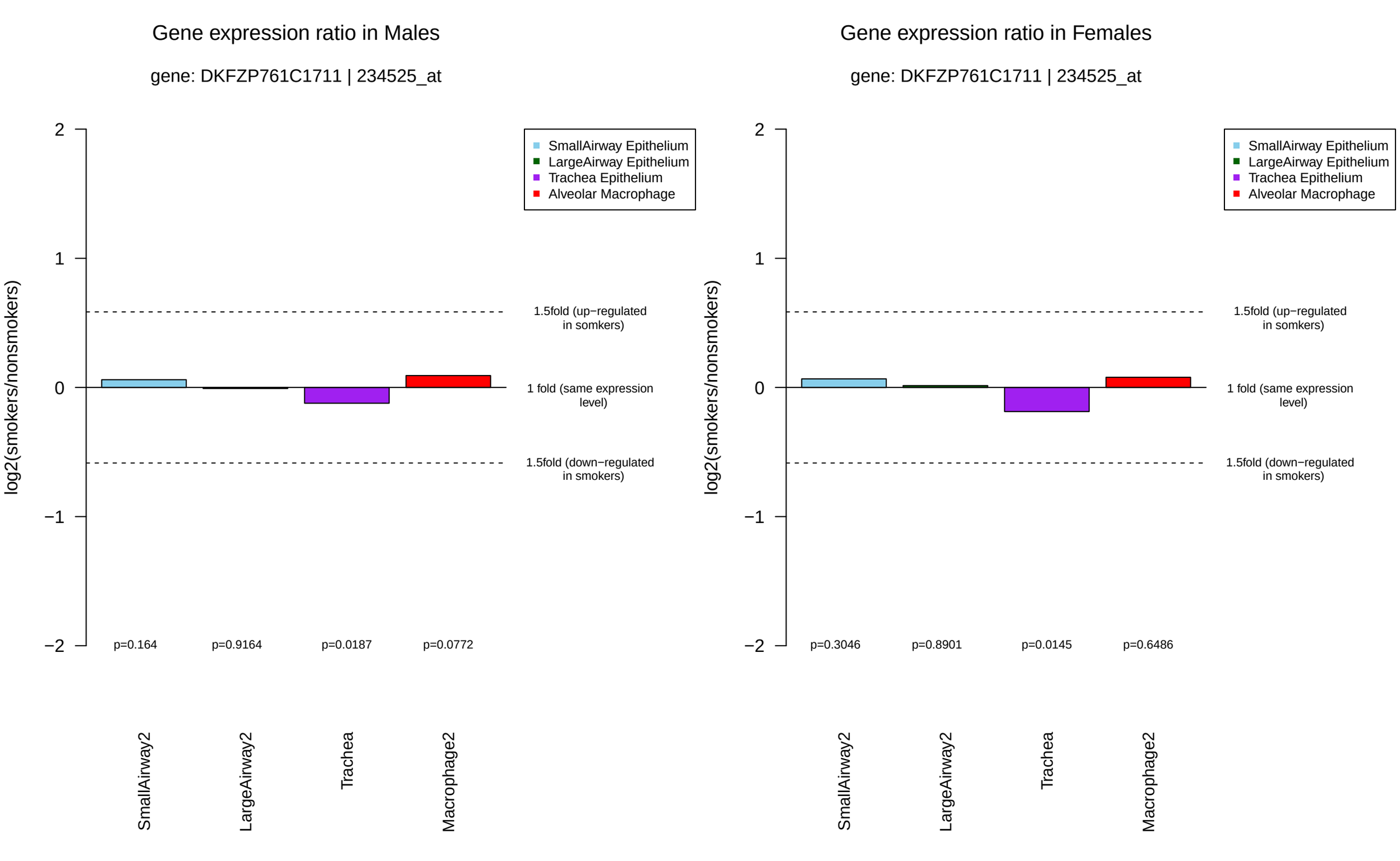
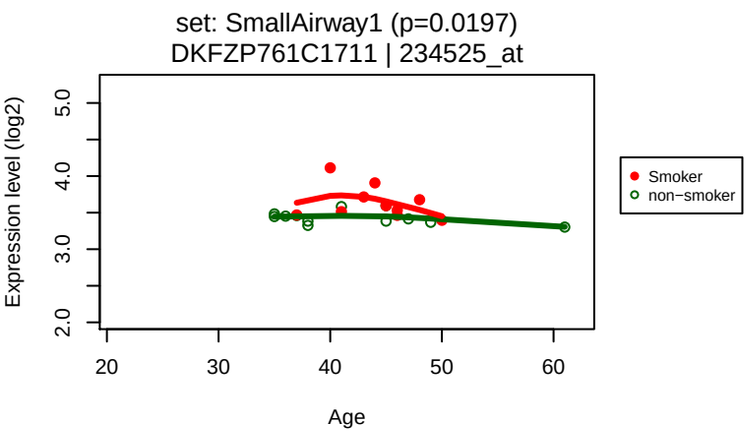
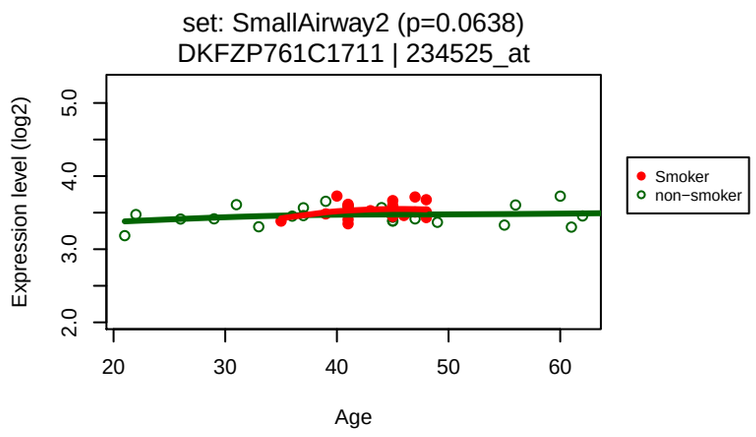
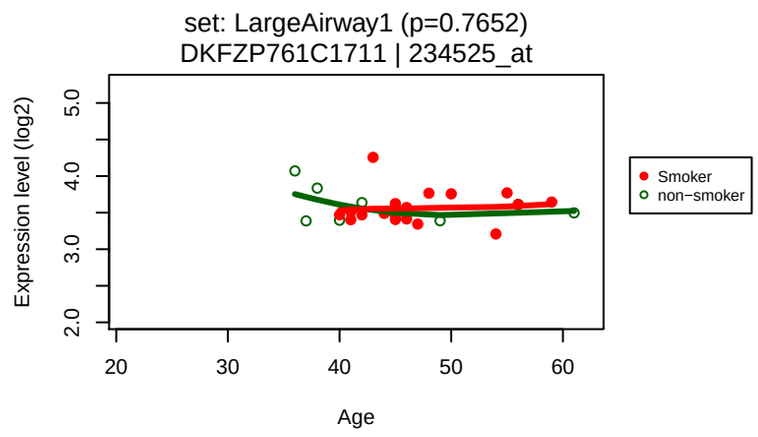

Gene expression of smokers and nonsmokers
gene: DKFZP761C1711 | 234525_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0197
p=0.0638
p=0.5505
p=0.7652
p=0.9694
p=0.005
p=0.7561
p=0.1307
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

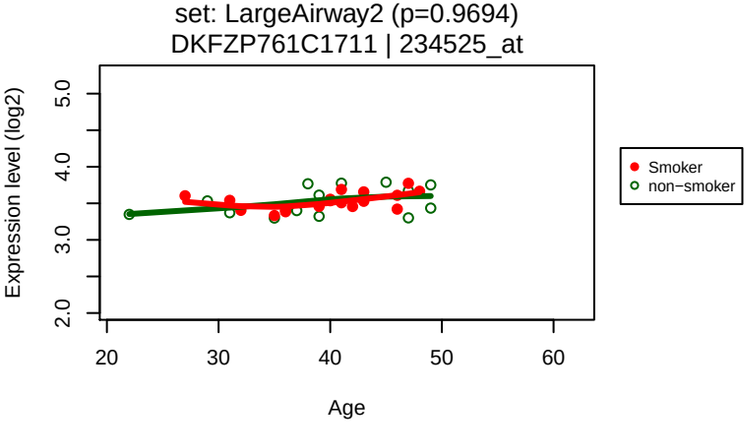

Gene expression ratio (smokers/nonsmokers)
gene: DKFZP761C1711 | 234525_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0197
p=0.0638
p=0.5505
p=0.7652
p=0.9694
p=0.005
p=0.7561
p=0.1307
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

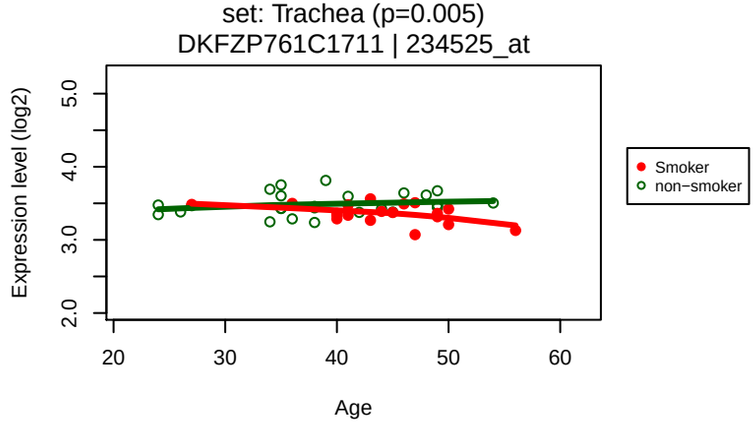

Gene expression ratio in Males
gene: DKFZP761C1711 | 234525_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.164
p=0.9164
p=0.0187
p=0.0772
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DKFZP761C1711 | 234525_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3046
p=0.8901
p=0.0145
p=0.6486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

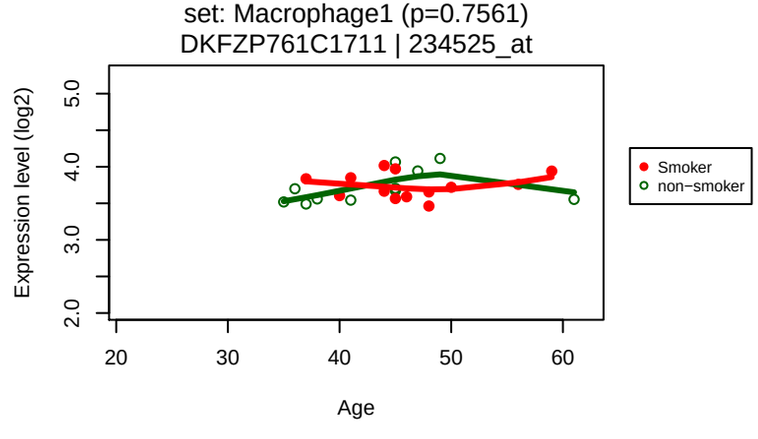

set: SmallAirway1 (p=0.0197)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

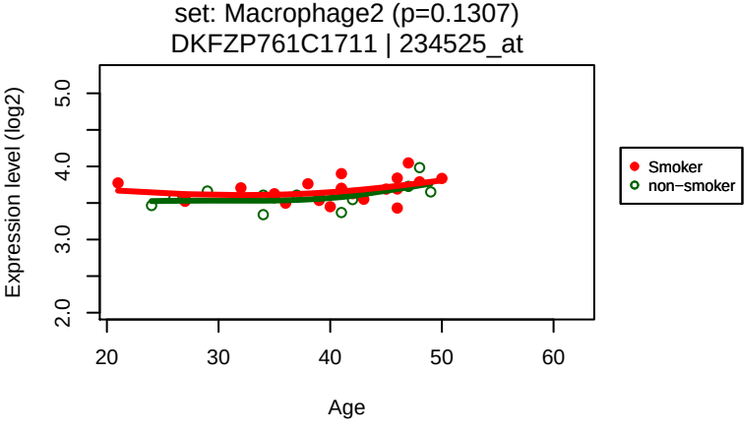

set: SmallAirway2 (p=0.0638)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

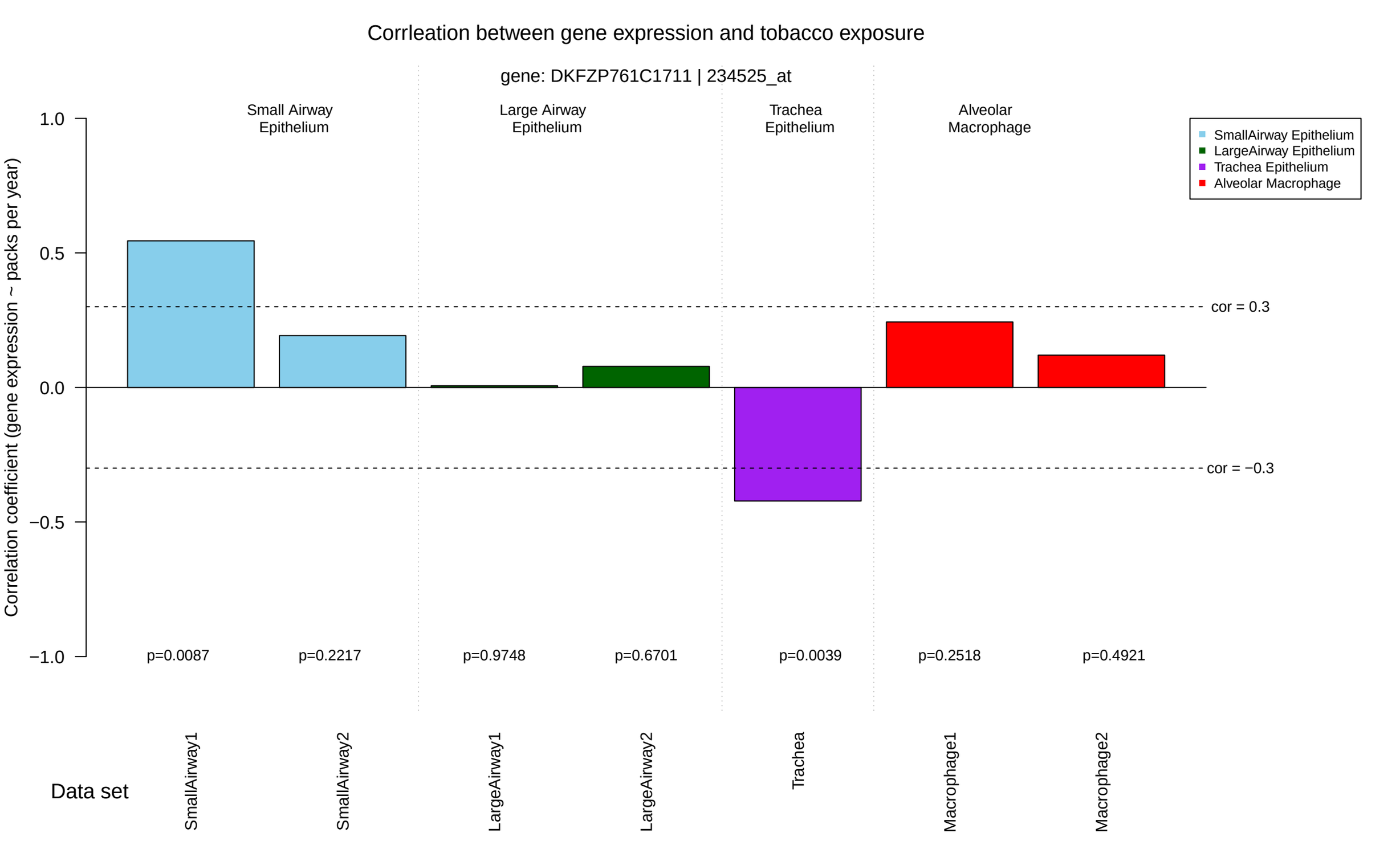

set: LargeAirway1 (p=0.7652)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9694)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.005)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7561)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1307)
DKFZP761C1711 | 234525_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DKFZP761C1711 | 234525_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0087
p=0.2217
p=0.9748
p=0.6701
p=0.0039
p=0.2518
p=0.4921
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2